

DS 3001 — A02 Question 1

Roshan Mahesh

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Setup

$$m(X) = \frac{1}{N} \sum_{i=1}^N x_i, \quad \text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y)), \quad s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

Part 1

Show that $m(a + bX) = a + b m(X)$.

$$\begin{aligned} m(a + bX) &= \frac{1}{N} \sum_{i=1}^N (a + bx_i) \\ &= \frac{1}{N} \left(\sum_{i=1}^N a + b \sum_{i=1}^N x_i \right) \\ &= \frac{1}{N} \left(Na + b \sum_{i=1}^N x_i \right) \\ &= a + b \cdot \frac{1}{N} \sum_{i=1}^N x_i \\ &= a + b m(X) \quad \blacksquare \end{aligned}$$

Part 2

Show that $\text{cov}(X, a + bY) = b \text{cov}(X, Y)$.

By Part 1, $m(a + bY) = a + b m(Y)$, so

$$(a + by_i) - m(a + bY) = (a + by_i) - (a + b m(Y)) = b(y_i - m(Y)).$$

Therefore

$$\begin{aligned}
\text{cov}(X, a + bY) &= \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) \left((a + by_i) - m(a + bY) \right) \\
&= \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) b(y_i - m(Y)) \\
&= b \cdot \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (y_i - m(Y)) \\
&= b \text{cov}(X, Y) \quad \blacksquare
\end{aligned}$$

Part 3

Show that $\text{cov}(a + bX, a + bX) = b^2 \text{cov}(X, X)$, and in particular that $\text{cov}(X, X) = s^2$.

$$\begin{aligned}
\text{cov}(a + bX, a + bX) &= \frac{1}{N} \sum_{i=1}^N \left((a + bx_i) - m(a + bX) \right)^2 \\
&= \frac{1}{N} \sum_{i=1}^N \left(b(x_i - m(X)) \right)^2 \\
&= b^2 \cdot \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2 \\
&= b^2 \text{cov}(X, X) \quad \blacksquare
\end{aligned}$$

Setting $Y = X$ in the definition of covariance,

$$\begin{aligned}
\text{cov}(X, X) &= \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) (x_i - m(X)) \\
&= \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2 \\
&= s^2 \quad \blacksquare
\end{aligned}$$

Combining the two results,

$$\text{cov}(a + bX, a + bX) = b^2 s^2.$$

Part 4

Consider a non-decreasing transformation $g(\cdot)$. Is it always true that $m(g(X)) = g(m(X))$?

No. Counterexample: let $g(x) = x^2$, which is non-decreasing on $[0, \infty)$, and take $N = 2$ with $x_1 = 1$ and $x_2 = 3$.

$$g(m(X)) = \left(\frac{1+3}{2}\right)^2 = 2^2 = 4$$

$$m(g(X)) = \frac{1^2 + 3^2}{2} = \frac{10}{2} = 5$$

$$5 \neq 4 \quad \blacksquare$$

More generally, equality holds for all X if and only if g is affine:

$$m(g(X)) = g(m(X)) \quad \forall X \quad \Longleftrightarrow \quad g(x) = a + bx.$$